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Dynamic partitioning shapes the in vivo organization of the *Escherichia coli* RNA degradosome

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The bacterial RNA degradosome is a central mediator of RNA turnover, yet how its components are organized and regulated in living cells remains unclear. Using live-cell single-molecule imaging, we quantified the spatial distribution and dynamics of the four major *Escherichia coli* degradosome components: RNase E, RhlB, PNPase, and enolase. These proteins occupied distinct membrane-associated and cytoplasmic pools whose relative abundance varied among proteins and changed with physiological state. In particular, PNPase underwent pronounced redistribution between membrane-associated and cytoplasmic states in response to altered RNA availability and growth conditions, whereas RNase E and RhlB remained largely membrane associated. To determine the functional consequences of this organization, we examined the degradation of *lacZ* reporter transcripts with different translation initiation strengths. PNPase and RhlB preferentially promoted degradation of weakly translated transcripts, whereas strongly translated transcripts were largely insensitive to their loss. Together, these results reveal that the bacterial RNA degradosome is not a static, uniformly assembled molecular machine. Instead, RNA decay is organized through dynamic intracellular partitioning of degradosome components, linking physiological state and translation status to transcript degradation.

RNA degradation | post-transcriptional gene regulation | single-molecule microscopy | biomolecular stoichiometry

Significance

RNA degradation is fundamental to bacterial gene regulation, yet degradosome organization within individual cells remains poorly understood. We show that the *Escherichia coli* degradosome is not a static molecular assembly but is organized through differential partitioning of its components between membrane-associated and cytoplasmic pools. Transcriptional arrest and the resulting depletion of cellular transcripts selectively remodel this organization. While RNase E and RhlB remain strongly membrane associated, PNPase becomes more membrane associated, and enolase shifts toward the cytoplasm. Partitioning also varies with growth conditions. Moreover, PNPase and RhlB preferentially promote degradation of weakly translated transcripts, linking degradosome organization to substrate-specific RNA decay. These findings suggest that dynamic, RNA-responsive degradosome organization helps bacterial cells regulate RNA degradation across physiological states.

RNA degradation is an important layer of bacterial gene regulation, controlling transcript abundance and contributing to stress responses and adaptation to environmental changes (1–3). As cellular conditions change, shifts in RNA synthesis and translation may impose different demands on RNA decay pathways. However, how bacterial RNA degradation machinery is organized and remodeled across physiological conditions remains unclear.

In bacteria, RNA decay is mediated by multienzyme degradosome assemblies, although their composition varies across species (4, 5). In *Escherichia coli*, the RNA degradosome (RNAD) is organized around the essential endoribonuclease RNase E (RNE), which initiates degradation of most mRNAs and processes multiple classes of RNA (6, 7). RNE functions as a scaffold through interactions mediated by its carboxy-terminal domain (CTD) (5, 7). The CTD recruits the ATP-dependent DEAD-box family RNA helicase RhlB, the 3'→5' exoribonuclease PNPase (PNP), and the glycolytic enzyme enolase (Eno) (6, 7). Together, these interactions are thought to organize RNA decay into a coordinated molecular machine.

RNAD is often depicted as a fully assembled molecular machine with fixed composition and stoichiometry (Fig. 1). Structural and biochemical studies have suggested that tetrameric RNE can simultaneously recruit multiple copies of RhlB, PNP, and Eno (4, 7–9). This model has provided a useful framework for understanding degradosome architecture and function (7). However, whether this model accurately describes the organization of RNA decay machinery in living cells remains uncertain.

Several observations suggest that RNAD composition may be more dynamic than the canonical model implies. Classic immunogold and freeze-fracture electron microscopy studies established that RNE-based degradosomes exist in vivo as multicomponent complexes associated with the inner membrane, while also suggesting that PNP and Eno can exist in non-scaffold-associated pools (8). Quantitative measurements further indicate that RNAD components are not produced in stoichiometric ratios corresponding to a fully occupied scaffold (14, 15). In addition, biochemical isolation of RNAD complexes has yielded widely varying estimates of component stoichiometry, with measured ratios depending strongly on purification conditions (11, 16, 17). These findings raise the possibility that substantial fractions of RNAD component proteins exist outside the canonical

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The authors declare no competing interest.

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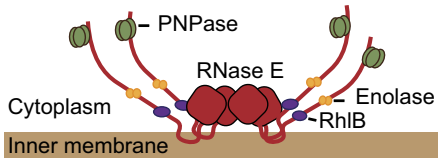


Fig. 1. Schematic of the canonical *E. coli* RNA degradosome. Membrane-anchored RNase E (RNE) forms a tetrameric scaffold that recruits the accessory factors RhlB, PNP, and Eno to assemble the canonical membrane-associated RNA degradosome. In this model, RNase E is a tetramer (10), and each RNE monomer interacts with a monomeric RhlB (7, 11), a trimeric PNP (12), and a dimeric Eno (13).

complex and that association with the RNE scaffold may be dynamically regulated rather than constitutive.

Recent advances in single-molecule imaging have begun to reveal the dynamic nature of bacterial RNA decay machineries. In *Bacillus subtilis*, proteins associated with the RNase Y-based degradosome exhibited multiple mobility states, and transcriptional inhibition altered both mobility and localization of those proteins (18). Such observations suggest that bacterial RNA decay machinery can undergo RNA-responsive reorganization. However, whether such reorganization reflects changes in scaffold composition, intracellular partitioning, or functional specialization of distinct component pools remains unclear.

In this study, we combined quantitative single-molecule imaging with functional measurements to define the in vivo organization of the *E. coli* RNAD. By resolving RNAD-associated and nonassociated populations within intact cells, we determined how RNA availability and growth condition shape degradosome composition and intracellular partitioning. Using *lacZ* reporters with different ribosome binding strengths, we further assessed whether RhlB and PNP preferentially promote decay of weakly translated and prematurely terminated RNAs. Together, these approaches enabled us to move beyond the prevailing fully assembled complex model and identify dynamic intracellular partitioning as an organizing principle of bacterial RNA decay.

Results

Single-Molecule Imaging Resolves Distinct Intracellular States of RNAD Component Proteins. To define the in vivo organization of the *E. coli* RNAD using single-molecule microscopy, we generated strains in which each of the four major RNAD components—RNE, RhlB, PNP, and Eno—was C-terminally fused to the photoconvertible fluorescent protein mEos3.2 (19) at its native chromosomal locus (*SI Appendix*, Tables S1 and S2). The functionality of the RNE-mEos3.2 fusion was established previously (20). The other three fusions, RhlB-mEos3.2, PNP-mEos3.2, and Eno-mEos3.2, were also functionally validated (*SI Appendix*, Fig. S1, Table S3, and *Supplementary Note 1*). Using the single-particle tracking workflow previously established for RNE-mEos3.2 (20), we measured the localization and diffusion of individual molecules in living cells. Under sparse photoconversion, each detected fluorescent spot was diffraction-limited and generally represented a single emitting mEos3.2-tagged protein (Fig. 2*A*), because multiple tagged molecules within the same complex were unlikely to fluoresce simultaneously. Accordingly, neither spot size nor intensity could be used to infer complex size.

In our previous study, we showed that RNE is almost exclusively membrane bound (20). Because the RNAD is anchored to the inner membrane through the membrane-targeting sequence (MTS) of RNE (21), and because RhlB, PNP, and Eno lack intrinsic membrane-binding regions (15, 22, 23), membrane localization of these accessory factors provides a readout of their association with the RNE scaffold. We therefore inferred RNAD association from their partitioning between membrane and cytoplasmic pools, with membrane localization indicating the RNAD-associated population and cytoplasmic localization indicating a nonassociated population. Furthermore, because a large membrane-associated degradosome complex is expected to diffuse slowly, we used diffusion as an independent and complementary measure of RNAD association.

RNAD Components Partition Differently between Membrane-Associated and Cytoplasmic Pools. Single-molecule localization was quantified using the position along the cell's short axis, normalized by cell width (xNorm; Fig. 2*B*). In this analysis, membrane-localized molecules are enriched near the cell periphery, whereas cytoplasmic molecules show a more central distribution. As reference standards for these two localization patterns, we imaged mEos3.2 fusions to lactose permease (LacY-mEos3.2) and β-galactosidase (β-Gal-mEos3.2), which exhibited the expected membrane-like and cytoplasmic xNorm histograms, respectively (Fig. 2 *C* and *D*). RNE and RhlB showed xNorm histograms very similar to that of LacY (Fig. 2 *E* and *F*), indicating that both proteins were nearly fully membrane bound under these conditions. In contrast, the xNorm histogram of PNP differed from that of the membrane reference LacY, consistent with a substantial cytoplasmic population in addition to a membrane-associated population (Fig. 2*G*). Eno showed an xNorm histogram similar to that of β-Gal (Fig. 2*H*), indicating that most Eno resided in the cytoplasm and was therefore not associated with RNAD. Thus, although all four proteins are considered canonical RNAD components, they occupy markedly different intracellular states in living cells.

We further quantified membrane association by fitting the xNorm histograms using the mathematical framework developed in our previous study (20). Model fitting was performed using a Markov chain Monte Carlo (MCMC) algorithm. The resulting membrane-bound fractions (MB%; Fig. 2*I* and *SI Appendix*, Fig. S2*A*) were consistent with the qualitative xNorm patterns. In particular, PNP showed an MB% of 68% [65, 72], and Eno showed an MB% of 5.6% [0.2, 14]. Brackets indicate 95% credible intervals from the MCMC posterior distribution, hereafter referred to as 95% CIs.

A common Slow-Diffusion State Defines RNAD In Vivo. We next analyzed the diffusion dynamics of RNAD components to determine whether they share a common mobility state corresponding to the membrane-associated degradosome. We first calculated diffusion coefficients of individual trajectories by fitting the mean-squared displacement (MSD) of each track to $MSD(\tau) = 4D\tau + b$, where D is the diffusion coefficient, τ is the lag time (multiples of 21.7 ms), and b accounts for static and dynamic localization errors (24, 25). While the preceding localization analysis included all trajectories containing at least four frames in length, reliable per-trajectory MSD fitting required at least 12 consecutive frames to provide sufficient lag points; the MSD analysis was therefore restricted to this longer-track subset. To characterize mobility across a broader trajectory set, we also performed a complementary analysis of one-frame squared displacements using all trajectories containing at least four frames. If multiple mobility populations are present, they can be resolved as distinct slow- and fast-diffusion components in the corresponding mobility distributions.

The D histograms of RNE and RhlB were well fit by a single Gaussian distribution with $R^2 > 0.97$ (Fig. 3 *A* and *B* and *SI Appendix*, Fig. S3*A* for two-population fits), with the peak values of 0.014 [0.013, 0.016] $\mu m^2/s$ for RNE and 0.014 [0.013,

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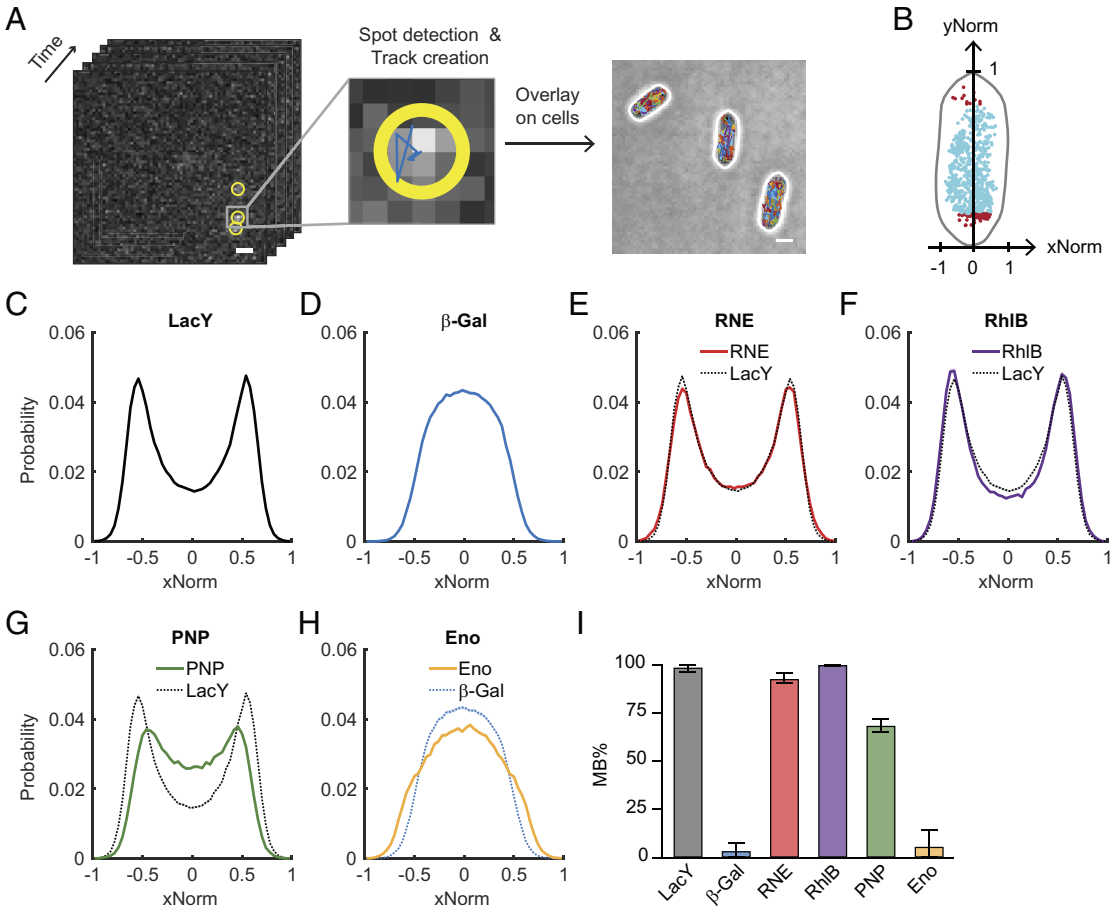


Fig. 2. Subcellular localization of RNAD component proteins. (*A*) Microscopy and analysis workflow. (Scale bar, 1 μm.) (*B*) Example cell showing the localization of Eno-mEos3.2 in normalized cell axes. Cyan spots in the cylindrical region of the cell were included in the xNorm histogram, while red spots in the endcaps were excluded. (*C*–*H*) xNorm histograms for proteins imaged in live cells, including LacY (*C*), β-Gal (*D*), RNE (*E*), RhlB (*F*), PNP (*G*), Eno (*H*). LacY and β-Gal reference histograms are overlaid for comparison. SEM estimated by bootstrapping is shown as a shaded area. (*I*) Membrane-binding percentage (MB%) of the RNAD components and control proteins. Error bars represent 95% CIs estimated from the MCMC posterior distribution. Data statistics are provided in *SI Appendix*, Table S6.

0.016] $\mu m^2/s$ for RhlB. This result is consistent with the earlier MB% analysis, which suggested that most cellular RNE and RhlB are membrane bound, and therefore RNAD associated.

In contrast, the D histograms of PNP and Eno contained pronounced long tails, indicating the presence of faster mobility populations (Fig. 3 *C* and *D* and *SI Appendix*, Fig. S3*A*). To separate these populations, the distributions were fit with a Gaussian mixture model, yielding a slow component (D_1) and fast components (D_2 and D_3). The slow components were close to those of RNE and RhlB with D_1 = 0.015 [0.015, 0.016] $\mu m^2/s$ for PNP and D_1 = 0.013 [0.013, 0.013] $\mu m^2/s$ for Eno. The fast components were D_2 = 0.044 [0.037, 0.051] $\mu m^2/s$ for PNP and D_2 = 0.033 [0.030, 0.037] $\mu m^2/s$ and D_3 = 0.19 [0.16, 0.22] $\mu m^2/s$ for Eno.

The common D_1 value (~0.014 $\mu m^2/s$) across all four components defines a shared slow-diffusion state, consistent with the membrane-associated RNAD complex (Fig. 3*E*). Notably, Eno exhibited this common slow state despite the weak membrane enrichment found in the localization analysis (Fig. 2*H*). Mobility-resolved xNorm distributions showed that the slow PNP and Eno trajectories were enriched at the membrane, whereas the fast trajectories were predominantly cytoplasmic (*SI Appendix*, Fig. S4), providing a mobility-based interpretation of the mixed membrane and cytoplasmic populations observed in the localization analysis (Fig. 2 *G* and *H*).

We note that the long-track MSD analysis emphasized molecules that remained trackable for longer periods. For Eno, restricting the xNorm analysis to this long-track subset increased its apparent membrane enrichment (*SI Appendix*, Fig. S5), indicating that rapidly moving cytoplasmic trajectories were underrepresented in the MSD analysis.

Next, we examined the survival distributions (1-CDF) of one-frame squared displacements using all trajectories containing at least four frames, with each trajectory contributing three squared displacements to ensure equal weighting (Fig. 3*F*). This analysis again recovered a common low-mobility state shared by all four RNAD components and resolved substantially faster populations of PNP and Eno that were underrepresented in the long-track MSD analysis (Fig. 3*G*). The fastest populations reached apparent diffusion coefficients of ~0.35 $\mu m^2/s$ for PNP and ~0.93 $\mu m^2/s$ for Eno. For comparison, the cytoplasmic reference β-Gal was dominated by a population with apparent diffusion coefficients of ~0.7 $\mu m^2/s$ (Fig. 3*G* and *SI Appendix*, Fig. S6*A*).

When localizations were binned according to the magnitude of the subsequent one-frame displacement, Eno localizations followed by large steps were predominantly cytoplasmic, whereas those followed by small steps occurred both at the membrane and in the cytoplasm (*SI Appendix*, Fig. S7*A*). Thus, most cellular Eno belongs to a highly mobile cytoplasmic population, while the low-mobility population includes both membrane-associated and cytoplasmic molecules. These results indicate that cytoplasmic Eno spans a broad range of mobilities that overlaps with that of membrane-associated Eno, such that the two spatial populations cannot be cleanly separated based on mobility alone.

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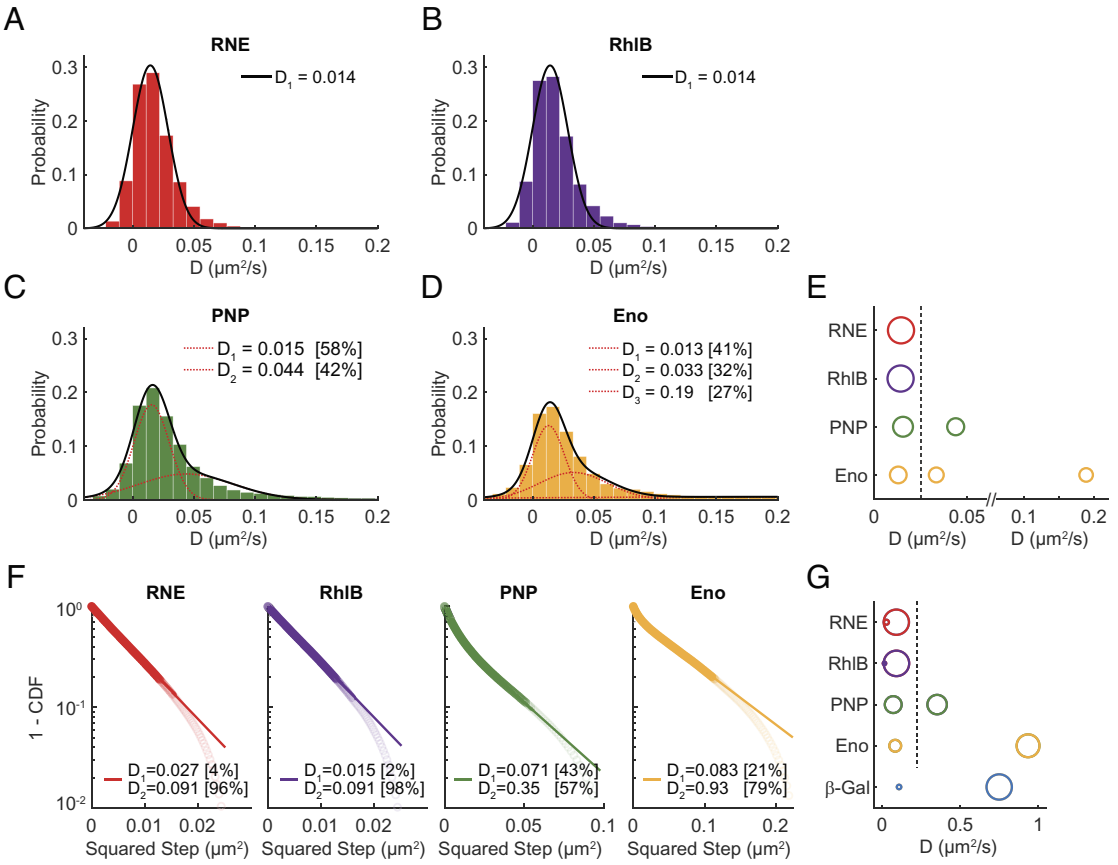


Fig. 3. Mobility states of RNAD components. (*A*–*D*) Distributions of diffusion coefficients obtained by MSD analysis for RNE (*A*), RhlB (*B*), PNP (*C*), and Eno (*D*). Only trajectories containing at least 12 consecutive frames were included. Histograms were fit using one Gaussian for RNE and RhlB, two for PNP, and three for Eno to reproduce the high-mobility tail of PNP and Eno. Dotted colored curves show the individual Gaussian components, and solid black curves show the corresponding summed fits. (*E*) Summary of mobility states obtained from the MSD analysis. The horizontal position of each circle indicates the fitted apparent diffusion coefficient (D_1, D_2, D_3), and its radius represents the corresponding population fraction. The x-axis is split to display the low- and high-mobility ranges. (*F*) Survival distributions (1-CDF) of one-frame squared displacements for RNE, RhlB, PNP, and Eno, using all trajectories containing at least four frames. Curves show two-population fits to the data shown as darker symbols, and the fitted apparent diffusion coefficients and population fractions are indicated in each plot. All four plots use the same logarithmic y-axis scale. (*G*) Summary of the mobility states obtained from the squared-displacement analysis for the four RNAD components and the cytoplasmic reference β-Gal. The full squared-displacement distribution for β-Gal is shown in *SI Appendix*, Fig. S6*A*. In *E* and *G*, the vertical dashed lines serve as visual guides to the RNAD-associated, low-mobility range based on the fitted RNE and RhlB mobility states within each analysis. Detailed fitting results and data statistics are provided in *SI Appendix*, Tables S5 and S6, respectively.

The per-trajectory MSD analysis provided trajectory-level estimates of mobilities, and the squared-displacement analysis captured a broader mobility distribution at the step level by including shorter trajectories. The apparent diffusion coefficients obtained from the two methods are not directly comparable because the methods use different trajectory subsets and treat localization uncertainty differently. In the MSD analysis, this contribution is largely captured by the fitted intercept b, but it contributes to the apparent diffusion coefficient in the one-frame squared-displacement analysis. Accordingly, the common low-mobility state appears at ~0.014 μm^2/s by MSD analysis and ~0.09 μm^2/s by squared-displacement analysis. The relationship between these estimates is discussed in *SI Appendix*, *Supplementary Note 2*. Importantly, both analyses support the same organizational principle: all four RNAD components share a common low-mobility state, whereas PNP and Eno additionally populate faster cytoplasmic states.

Membrane-Associated Pools Reveal Near-Canonical RNAD Stoichiometry under the Reference Condition. The differential partitioning of RNAD components raised the question of how closely the membrane-associated population resembles the canonical degradosome complex in vivo. To estimate RNAD stoichiometry under the reference condition, we combined the membrane-bound fractions measured by live-cell xNorm analysis with published protein abundance values for *E. coli* derived from ribosome profiling (14), while noting that these abundance measurements were obtained under different growth conditions.

Because ribosome profiling reports synthesis of individual polypeptide chains, the published abundance values correspond to monomeric subunits. We therefore first converted these values to functional oligomeric forms for tetrameric RNE (RNE_4), trimeric PNP (PNP_3), and dimeric Eno (Eno_2). We then multiplied them by the corresponding MB% estimated from xNorm histograms (Fig. 2*I*). This yielded an ensemble-average membrane-associated ratio of approximately 1 RNE_4: 4-5 RhlB: 3-4 PNP_3: 2 Eno_2 (Table 1). Thus, RhlB and PNP were present in the membrane-associated pool at levels broadly consistent with the canonical expectation for a fully assembled RNAD complex, 1 RNE_4: 4 RhlB: 4 PNP_3: 4 Eno_2 (Fig. 1), while Eno appeared lower.

One caveat is that our MB% values were measured in cells grown in M9 minimal medium supplemented with 0.2% glycerol, casamino acids, and thiamine (M9 gly + CAAT) at 30 °C, whereas the protein abundance values were obtained from cells grown in Neidhardt EZ rich defined medium or MOPS medium with 2% glucose at 37 °C. Despite this difference in growth conditions, the inferred membrane-associated stoichiometry places RhlB and

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Table 1. Estimated membrane (MB)-associated stoichiometry of RNAD components

		Abundance from EZRDM		Abundance from MOPS glucose	
	MB%	Multimer	MB-associated multimers, relative to RNE	Multimer	MB-associated multimers, relative to RNE
RNE_4	92.6 ± 1.2	682	1	265	1
RhlB	99.8 ± 0.2	3,459	5.5 ± 0.1	1,055	4.3 ± 0.1
PNP_3	68.3 ± 1.9	3,968	4.3 ± 0.1	1,181	3.3 ± 0.1
Eno_2	5.6 ± 3.8	27,111	2.4 ± 1.6	10,617	2.4 ± 1.7

Published protein abundance values derived from ribosome profiling (14) were converted to functional oligomeric forms and multiplied by the corresponding membrane-bound fraction (MB%) measured from xNorm histogram fitting. MB% values are shown as the mean ± SD of the MCMC posterior distribution. MB-associated abundances were normalized to RNE, and errors represent uncertainty propagated from MB% estimation.

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PNP broadly near the canonical expectation. Eno appeared lower than the canonical value of 4, although this estimate is more sensitive to uncertainty in MB% because of the large cellular abundance of Eno. Indeed, the uncertainty range arising from MB% estimation spans the canonical value for Eno.

Together, this analysis reconciles the canonical degradosome model with the single-cell partitioning data by showing that a near-canonical membrane-associated complex can coexist with large nonassociated pools of specific accessory factors.

RNA Depletion Selectively Remodels PNPase and Enolase Localization and Mobility. We next asked whether this partitioning is regulated by RNA availability. To reduce cellular mRNA levels, cells were treated with rifampicin (rif), an inhibitor of transcription initiation (26), for 15 min before imaging, and were imaged within 1 h in the continued presence of rif on the agarose pad. Because most *E. coli* mRNAs have half-lives shorter than 10 min (27), the 15-min pretreatment is expected to substantially reduce the cellular mRNA pool by the start of imaging. We also imaged LacY and β-Gal under the same conditions as reference proteins for 100% and 0% membrane binding, respectively, to account for any rif-induced changes in xNorm distributions (Fig. 4 *A* and *B*).

Surprisingly, the RNAD components exhibited different localization responses upon rif treatment (Fig. 4 *C–F* and *SI Appendix*, Figs. S2*B* and S8*A*). Notably, RNE and RhlB were largely unaffected by rif treatment and remained predominantly membrane localized, whereas PNP became more membrane-associated and Eno became more cytoplasmic. Quantification of MB% by xNorm fitting confirmed these trends. PNP displayed a marked increase in MB%, from 68 to 92%, indicating a reduction in its cytoplasmic pool (Fig. 4*G*). The rif-induced increase in membrane association of PNP was also evident in two-dimensional localization histograms (*SI Appendix*, Fig. S9), in which a central depletion region became more apparent with higher MB%.

To determine whether these localization changes were accompanied by altered mobility, we analyzed the rif-treated datasets using the same per-trajectory MSD and squared-displacement approaches described above (*SI Appendix*, Figs. S3*B* and S6*B*). Both analyses recovered a common low-mobility state shared by all four RNAD components after rif treatment, with apparent diffusion coefficients generally higher than those of the corresponding states in untreated cells (Fig. 4 *H* and *I*). The fraction of PNP assigned to the low-mobility state increased after rif treatment, paralleling its increased membrane association and supporting greater partitioning of PNP into the membrane-associated RNAD population.

In contrast, Eno mobility increased substantially following rif treatment. The squared-displacement analysis resolved a fast Eno population with an apparent diffusion coefficient of ~2.0 $\mu m^2/s$ (Fig. 4*I*). Using the same analysis, β-Gal showed no detectable rif-dependent mobility increase, suggesting that a global change in cytoplasmic mobility is unlikely to fully explain the Eno response. Mobility-resolved xNorm analysis further showed that a subset of the remaining low-mobility Eno population was membrane associated (*SI Appendix*, Fig. S7*B*), consistent with continued association of a minor Eno population with membrane-bound RNE.

The rif-induced spatial redistribution also altered the inferred composition of the membrane-associated RNAD pool. Assuming that total protein abundance does not change substantially during the short rif-treatment period, we applied the MB% values measured after rif treatment to the same abundance-based framework used above. This analysis indicated that the inferred PNP occupancy increased, reaching approximately 5.4 PNP_3 per RNE_4 when combined with abundance values from EZRDM and 4.2 PNP_3 per RNE_4 when combined with abundance values from MOPS glucose (Table 2). In contrast, Eno was strongly depleted from the membrane-associated pool, with an inferred value of approximately 0.3 Eno_2 per RNE_4 in both abundance datasets.

Together, these findings reveal an RNA-dependent reorganization of both the spatial partitioning and mobility of specific RNAD components. The common low-mobility RNAD state persisted after RNA depletion, but PNP shifted toward this state, whereas Eno became more cytoplasmic and substantially more mobile. One possible explanation is that abundant cytoplasmic RNA promotes retention of PNP in the cytoplasm, whereas RNA depletion after rif treatment shifts the equilibrium toward association with the membrane-bound RNE scaffold.

RNAD Organization Shifts in Succinate-Grown Cells. Cellular RNA abundance and transcriptional flux vary substantially across growth conditions (28, 29). We therefore asked whether RNAD organization can be modulated under distinct physiological states. To test this, we analyzed diffusion and localization of RNAD components in cells grown in M9 succinate medium without CAAT. Succinate is a poorer carbon source than glycerol under our conditions, with doubling times of ~350 min for MG1655 grown in M9 succinate and 82 min in M9 gly + CAAT at 30 °C.

Single-molecule tracking revealed diffusion behavior qualitatively similar to that observed in M9 gly + CAAT. In the MSD analysis, all four RNAD components shared common slow-diffusion states with apparent diffusion coefficient of ~0.02 and 0.05 $\mu m^2/s$, and PNP and Eno exhibited broader distributions extending to higher mobilities (*SI Appendix*, Figs. S3*C* and S10*A*). The common slow-mobility states were 1.5- to 2-fold faster than those measured in M9 gly + CAAT, indicating increased mobility of the RNAD-associated state in cells grown in M9 succinate. The fast-diffusing populations of PNP and Eno were also shifted to higher values relative to M9 gly + CAAT (*SI Appendix*, Fig. S10*A*). A complementary squared-displacement analysis that included shorter trajectories recovered the same overall organization (*SI Appendix*, Figs. S6*C* and S10*B*).

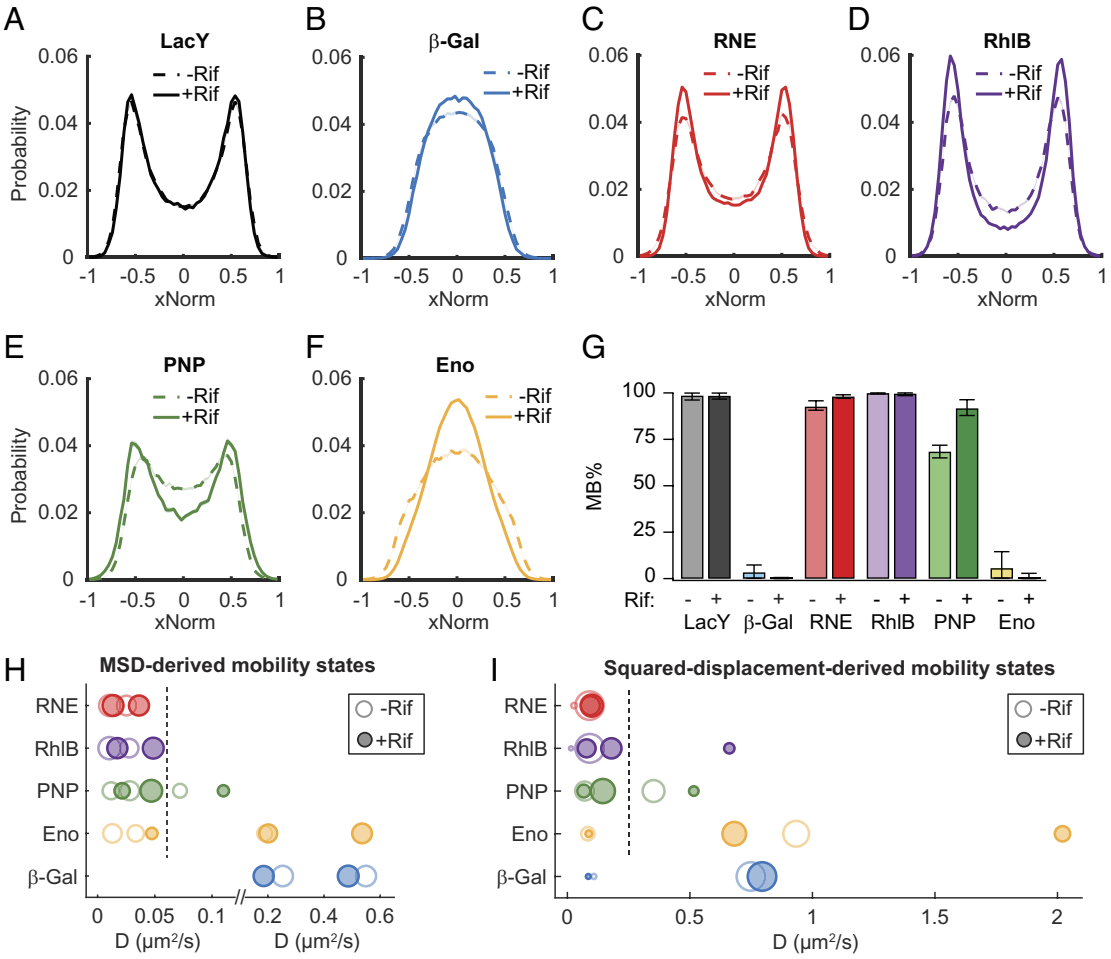


Fig. 4. RNAD component localization and mobility in rifampicin-treated cells. (*A* and *B*) Reference proteins: LacY, a membrane protein, and β-Gal, a cytoplasmic protein. (*C*–*F*) The four canonical RNAD components. Shaded colored regions around the xNorm histogram lines represent the SEM estimated by bootstrapping. (*G*) Membrane-binding percentage (MB%) of the RNAD components and control proteins before and after rifampicin treatment. Error bars represent 95% CIs estimated from the MCMC posterior distribution. (*H*) Summary of mobility states obtained from per-trajectory MSD analysis for the four RNAD components and the cytoplasmic reference β-Gal before and after rif treatment. The horizontal position of each circle indicates the fitted apparent diffusion coefficient, and circle radius represents the corresponding fitted population fraction. Open and filled circles indicate untreated and rif-treated cells, respectively. The x-axis is split to display the low- and high-mobility ranges. (*I*) Corresponding summary of mobility states obtained from the one-frame squared-displacement analysis. In *H* and *I*, the vertical dashed lines serve as visual guides to the RNAD-associated low-mobility range based on the fitted RNE and RhlB mobility states within each analysis. Detailed data statistics are reported in *SI Appendix*, Table S6.

We next examined xNorm histograms. Most proteins, including LacY and β-Gal, showed little change relative to M9 gly + CAAT. PNP shifted toward a slightly higher MB%, whereas Eno shifted toward a lower MB% (Fig. 5 and *SI Appendix*, Figs. S2*C* and S8*B*), a trend qualitatively similar to that observed after rif treatment (Fig. 4 *E* and *F*).

These observations indicate that RNAD organization is not constant across growth conditions. Instead, both the localization and mobility of degradosome components are remodeled by physiological state. One possible interpretation is that these changes reflect altered partitioning of RNAD components in response to differences in RNA availability and carbon metabolism across growth conditions. Because glycerol and succinate enter different points in central carbon metabolism (30), enolase partitioning may be linked to changes in metabolic state, potentially including altered metabolic flux and increased coupling between glycolytic and RNA decay pathways under carbon-limited conditions (31). Also, these observations raise the possibility that enolase

Table 2. Estimated membrane (MB)-associated stoichiometry of RNAD components in rifampicin-treated cells

		Abundance from EZRDM		Abundance from MOPS glucose	
	MB%	Multimer	MB-associated multimers, relative to RNE	Multimer	MB-associated multimers, relative to RNE
RNE_4	98.0 ± 0.5	682	1	265	1
RhlB	99.6 ± 0.4	3,459	5.2 ± 0.0	1,055	4.0 ± 0.0
PNP_3	91.7 ± 2.7	3,968	5.4 ± 0.1	1,181	4.2 ± 0.1
Eno_2	0.7 ± 0.7	27,111	0.3 ± 0.3	10,617	0.3 ± 0.3

As in Table 1, published protein abundance values from ribosome profiling (14) were converted to functional oligomeric forms and multiplied by the membrane-bound fraction (MB%) measured from rif-treated cells. MB% values are shown as the mean ± SD of the MCMC posterior distribution. MB-associated abundances were normalized to RNE, and errors represent uncertainty propagated from MB% estimation.

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function may depend in part on its partitioning between cytoplasmic and membrane-associated RNAD pools.

RhlB and PNP Preferentially Promote Decay of Weakly Translated *lacZ* Transcripts. The RNA-dependent redistribution of PNP led us to a functional question: Do RNAD components contribute differently to decay of distinct classes of RNA substrates? In our recent work, we showed that translation initiation rate, altered by ribosome binding site (RBS) strength, influences the abundance of prematurely released transcripts arising from disrupted transcription–translation coupling (32). Based on this, we asked whether RhlB and PNP contribute more strongly to the decay of weakly translated mRNAs than of efficiently translated transcripts.

To test this idea, we examined the degradation kinetics of *lacZ* mRNA carrying either a strong (native) or weak RBS (Fig. 6*A*). In our previous work, 75-s transient induction of *lacZ* and time-course quantification of *lacZ* 5' levels by RT-qPCR resolved two decay phases: k_{d1}, reflecting decay of nascent or prematurely released transcripts, and k_{d2}, corresponding to decay of fully transcribed transcripts (32).

lacZ transcripts bearing the native RBS exhibited negligible k_{d1}, indicating minimal cotranscriptional decay, minimal premature release, and predominantly posttranscriptional decay (Fig. 6*B*). To determine whether RhlB and PNP contribute to degradation of these transcripts, we measured *lacZ* mRNA decay in wild-type (WT), Δ*rhlB*, and Δ*pnp* strains. For the native-RBS transcript, k_{d1} remained close to zero, and the major decay phase (k_{d2}) was indistinguishable among WT, Δ*rhlB*, and Δ*pnp*, consistent with RNE-mediated cleavage remaining rate-limiting for this transcript (Fig. 6 *C* and *D*).

In contrast, *lacZ* transcripts with a weak RBS displayed an elevated k_{d1}, consistent with increased production of prematurely released RNA species, whose 5' region was degraded before accumulation of the bulk *lacZ* 3' signal (Fig. 6*E*). In Δ*rhlB* and Δ*pnp* strains, relative RNA levels declined more slowly than in WT during both in the k_{d1} and k_{d2} fitting regions (Fig. 6*F*), resulting in reductions in both degradation rates (Fig. 6*G*). k_{d1} was particularly sensitive to the loss of RhlB and PNP (Fig. 6*G*). These results indicate that RhlB and PNP contribute to efficient degradation of weakly translated or nontranslating transcripts, whereas decay of strongly translated transcripts is less dependent on these accessory components.

Discussion

By combining live-cell single-molecule localization, diffusion measurements, and functional RNA decay assays, we examined how the major components of the *E. coli* RNA degradosome (RNAD) are organized in vivo. Our results support a view in which RNAD is a membrane-anchored but dynamically partitioned system composed of component-specific membrane-associated and cytoplasmic pools whose balance shifts with physiological state.

This view helps bridge a long-standing gap between biochemical models of the RNAD and its behavior in living cells. Earlier studies based on affinity purification yielded differing pictures of accessory-protein occupancy, with some suggesting overrepresentation of RhlB (11, 16) and Eno (16) and underrepresentation of PNP (11, 16, 17), relative to canonical stoichiometry. Our approach does not resolve the composition of individual complexes directly, but it does provide an in situ readout of how RNAD components partition between membrane-associated and cytoplasmic pools. At the ensemble level, the membrane-associated pool exhibits near-canonical stoichiometry even though substantial fractions of PNP and Eno remain outside this pool, helping reconcile canonical biochemical models with the organization observed in living cells.

Among the accessory factors, Eno and PNP showed the clearest evidence of dynamic partitioning, although they responded in opposite directions to RNA depletion. Eno's MB% decreased from

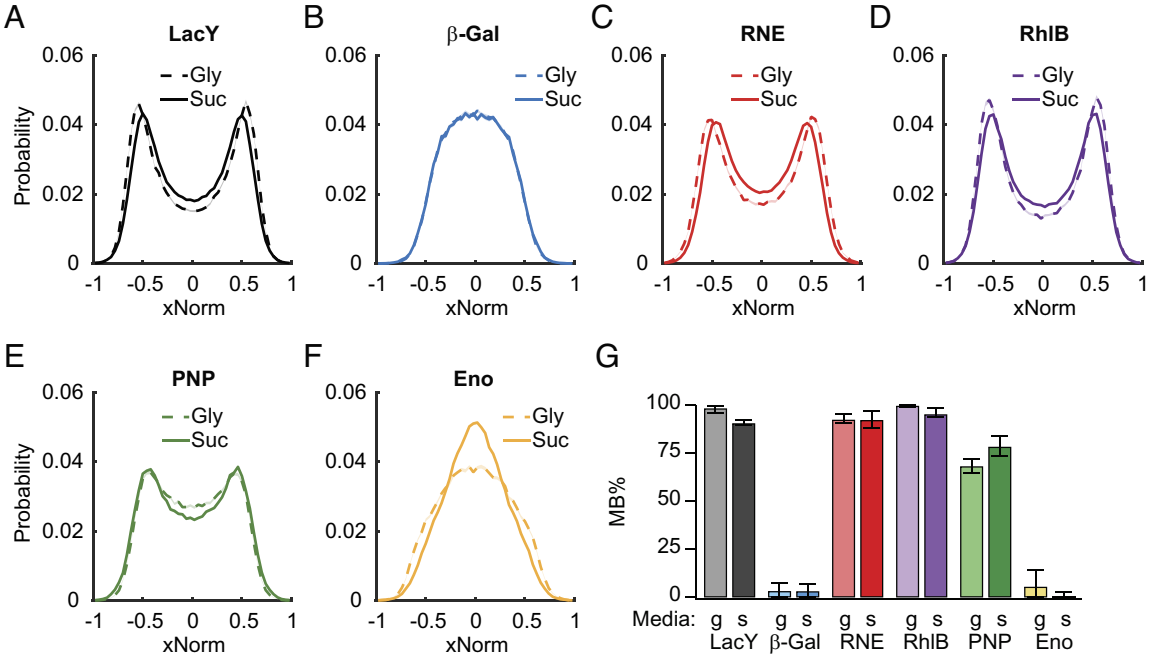


Fig. 5. Subcellular localization of RNAD components in cells grown in M9 succinate. (*A*–*F*) xNorm histograms of proteins in cells grown in M9 succinate, including reference proteins LacY (*A*) and β-Gal (*B*) as well as the four RNAD component proteins, RNE (*C*), RhlB (*D*), PNP (*E*), and Eno (*F*). Shaded colored regions around the xNorm histogram lines represent the SEM from bootstrapping. Data from M9 glycerol + CAAT are drawn for a comparison. (*G*) MB% quantified from the xNorm histograms shown in (*A*–*F*). "g" represents M9 glycerol +CAAT and "s" represents M9 succinate. Error bars represent 95% CIs estimated from the MCMC posterior distribution. Data statistics are listed in *SI Appendix*, Table S6.

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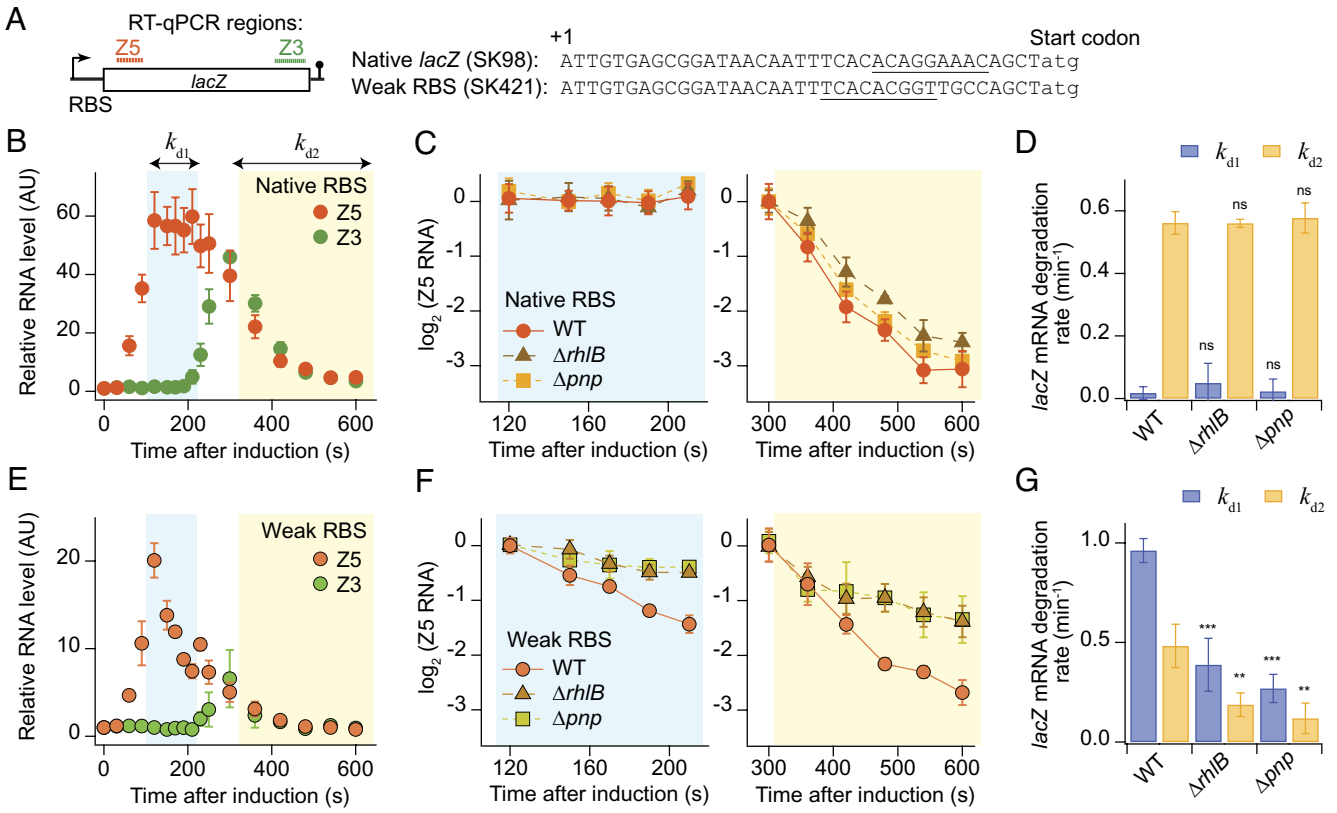


Fig. 6. mRNA degradation analysis using the *lacZ* reporter in wild-type, Δ*rhlB*, and Δ*pnp* strains. (*A*) Schematic of the *lacZ* reporter at its native chromosomal locus with different ribosome-binding-site (RBS) sequences. DNA sequences are shown from the transcription start site (+1) to the start codon (atg). Underlined sequences are Shine–Dalgarno elements estimated by an RBS calculator (33). The 5′ and 3′ RT-qPCR amplicons are indicated as Z5 and Z3, respectively. (*B*) Time courses of RNA abundance measured at the Z5 and Z3 regions of *lacZ* in the native-RBS reporter (SK98). *lacZ* transcription was induced with 0.2 mM IPTG at t = 0 and rerepressed by addition of 500 mM glucose at t = 75 s. Blue and yellow shaded regions indicate the time windows used to quantify the Z5 decay rates, k_{d1} and k_{d2}, respectively, by exponential fitting. (*C*) Normalized decay kinetics of Z5 RNA within the blue and yellow fitting windows in SK98 (WT), SK301 (Δ*rhlB*), and SK306 (Δ*pnp*). RNA levels were normalized to the first time point shown in each graph. (*D*) Corresponding fitted k_{d1} and k_{d2} values. (*E*–*G*) Corresponding analysis of the weak-RBS reporter, including time course data in WT (SK421, *E*), normalized decay kinetics in SK421 (WT), SK437 (Δ*rhlB*), and SK431 (Δ*pnp*, *F*), and fitted values (*G*). The induction and repression conditions and fitting windows were the same as in *B*. The WT data for native- and weak-RBS reporters (SK98 and SK421, respectively) were previously published in ref. 32. Data are mean ± SD of four (SK98, SK421, SK301, SK437) and three (SK306, SK431) biological replicates. In panel *D*, for k_{d1}, $P = 0.36$ (SK301), $P = 0.83$ (SK306); for k_{d2}, $P = 0.95$ (SK301), $P = 0.64$ (SK306), from two-sided two-sample *t* tests relative to SK98. In panel *G*, for k_{d1}, ***$P = 0.0010$ (SK437), ***$P = 0.00020$ (SK431); for k_{d2}, **$P = 0.0056$ (SK437), **$P = 0.0092$ (SK431), from two-sided two-sample *t* tests relative to SK421.

5.5 to 0.73% after rif treatment, indicating dissociation from RNAD. However, it does not imply complete dissociation of Eno from membrane-bound RNE. A slowly diffusing Eno population remained detectable after rif treatment, with a diffusion coefficient comparable to that of RNE under the same condition, and its mobility-resolved xNorm distribution retained a membrane-associated population. Thus, a subset of Eno may remain associated with RNE even as most Eno shifts to the cytoplasm. Rif also caused a pronounced increase in Eno mobility, a response not observed for the cytoplasmic control β-Gal, suggesting that it is not simply a general increase in cytoplasmic protein mobility (34). Because *E. coli* enolase has been recovered as an RNA-associated protein (35), this increase may instead reflect loss of RNA-dependent interactions, although direct Eno-RNA binding and its functional significance remain unclear.

PNP, in contrast, had a substantial cytoplasmic population in untreated cells but became more membrane-associated after rif treatment and in succinate-grown cells. One plausible interpretation is that abundant cytoplasmic RNA favors RNA-bound pools of PNP, whereas RNA depletion shifts the equilibrium toward PNP association with the membrane-bound RNE scaffold. In this view, PNP may interact with RNA in both scaffold-associated and nonassociated states, with the balance between these pools set by cellular RNA availability. Previous biochemical analysis also identified PNP as a particularly condition-responsive RNAD component. The PNP:RNE ratio increased nearly twofold in cells grown at 16 °C or treated with α-methylglucoside to induce phosphosugar stress (11). Although this biochemical measurement is distinct from the live-cell partitioning analyzed here, both studies identify PNP as an especially dynamic component across physiological states. Such condition dependence may also explain why biochemical studies have not converged on a single degradosome stoichiometry in the literature. If a substantial fraction of PNP is conditionally associated with RNE, its apparent occupancy will depend not only on growth condition but also on how complexes are extracted or stabilized.

A possible functional consequence of PNP partitioning is a division of labor between membrane-associated decay initiation and downstream processing of RNA fragments. RNE is anchored to the inner membrane and initiates RNA decay by endonucleolytic cleavage. PNP, in contrast, partitions between the membrane-associated scaffold and the cytoplasm, allowing it to act on RNA decay intermediates either locally at the RNAD or after their release from the scaffold (Fig. 7). Maintaining a substantial cytoplasmic pool of PNP could therefore facilitate clearance of decay intermediates without requiring all processing to occur at the membrane-bound scaffold. At the same time, scaffold association of PNP likely remains important for coordinated decay of specific substrates, enabling rapid local handoff after RNE-mediated cleavage (36). This view is consistent with recent 3′-end profiling

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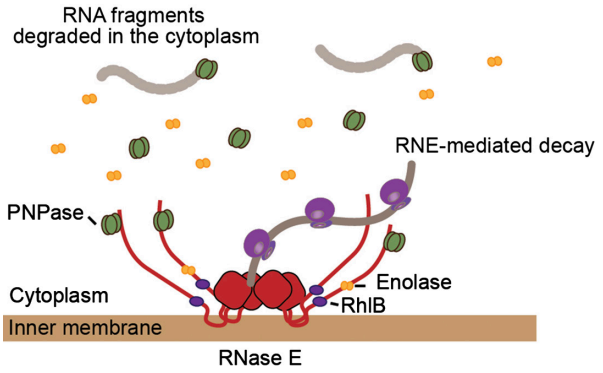


Fig. 7. Model for dynamic partitioning and functional organization of RNAD components in *E. coli*. RNE is anchored to the inner membrane and serves as a scaffold for assembly of the RNAD. RhlB is largely associated with the membrane-bound scaffold, whereas PNP partitions between membrane-associated and cytoplasmic pools in an RNA-dependent manner. Eno is largely cytoplasmic, with only a smaller fraction associated with the membrane-bound complex. In this model, membrane-associated RNE initiates RNA decay by endonucleolytic cleavage, after which RNA fragments can be further degraded by PNP either at the scaffold or in the cytoplasm. Gray lines represent RNA.

showing that most *E. coli* mRNAs are decay intermediates rather than full-length transcripts, that these intermediates are depleted from ribosome-rich fractions, and that they accumulate further in Δ*pnp* cells (37). It is also consistent with transcriptome-wide studies showing that different classes of *E. coli* mRNAs respond differently to mutations in RNAD components, suggesting that RNA decay need not proceed through a single uniformly assembled degradosome pathway (38).

The *lacZ* decay experiments add an important functional dimension to this model. Loss of *rhlB* or *pnp* had little effect on degradation of strongly translated *lacZ* transcripts, but both deletions slowed decay of weakly translated transcripts. In particular, k_{d1}, reflecting decay of prematurely terminated transcripts (32), showed notable sensitivity. Thus, RhlB and PNP matter most when translation is weak and transcripts are likely to be prematurely released or less covered by ribosomes. Although these experiments do not directly assign the degradation of these RNA substrates to cytoplasmic and membrane-associated PNP pools, they support the broader idea that degradosome factors contribute unequally to RNA decay across substrate classes. Together with the rif-dependent redistribution of PNP, our findings suggest a potential coupling between substrate selectivity and PNP partitioning. Weakly translated and prematurely terminated RNAs, as well as decay intermediates generated from them, may contribute to a cytoplasmic RNA substrate pool that engages PNP during normal growth. Following transcription arrest, broad depletion of cellular mRNAs and their decay intermediates could reduce this substrate pool and shift PNP toward the membrane-bound RNE scaffold. The increased membrane association of PNP after rif treatment therefore suggests that the cytoplasmic PNP population during normal growth is functionally engaged with RNA rather than simply representing stoichiometric excess.

Several caveats are worth keeping in mind. Our stoichiometry estimates combine measured MB% values with published abundance data rather than direct copy-number measurements under the same growth conditions. In addition, the relevant oligomeric state of RhlB in vivo remains uncertain. We assumed monomeric RhlB interacting with one RNE monomer based on refs. 7 and 11; however, other studies have suggested that RhlB may function as a dimer (39, 40). If a dimeric RhlB interacts with each RNE monomer, the inferred RhlB occupancy would be reduced (i.e., under occupied). These limitations affect the precision of the stoichiometric estimates, but not the central conclusions from the imaging data: RhlB is strongly membrane associated, PNP is dynamically partitioned, and Eno is only weakly represented in the membrane-associated pool.

Overall, our results support a view of the *E. coli* RNA degradosome as a membrane-anchored but compositionally flexible RNA decay system. Rather than existing as one static assembly, it appears to comprise component-specific associated and nonassociated populations whose balance shifts with RNA availability and growth condition. More broadly, our findings suggest that bacterial cells can regulate RNA decay not only through enzyme activity or substrate recognition, but also by redistributing decay factors between distinct intracellular pools. Given that RNE serves as a scaffold for RNA degradosome assembly across bacteria and chloroplasts (5), dynamic intracellular partitioning may represent a broader principle for organizing RNA decay machinery in vivo.

Materials and Methods

Cloning and Strain Construction. Strains used in this study are listed in *SI Appendix*, Table S1 with the details of construction listed in *SI Appendix*, Table S2. Doubling times were measured as described previously (20). Briefly, cells were grown overnight in M9 glycerol +CAAT or in M9 succinate at 30 °C, diluted 10^4-fold, and the optical density at 600 nm (OD_{600}) was measured using a microplate reader (Synergy HTX multimode reader, BioTek), with continuous shaking. Data analysis included fitting the background-subtracted data to an exponential function within OD_{600} between 0.001 and 0.12 (*SI Appendix*, Table S3).

Cell Growth and Sample Preparation. Cells were grown in M9 minimal media supplemented with 0.2% (v/v) glycerol (Invitrogen 15514011), 0.1% (w/v) casamino acids (Bacto 223050), and 1 μg/mL thiamine (Research Products International T21020) at 30 °C with shaking at 220 rpm. Starter cultures were inoculated from several colonies and grown for 6 to 8 h, then diluted ≥3,000-fold and grown overnight to early exponential phase (OD_{600} = 0.15 to 0.25) prior to imaging.

For M9 succinate conditions, cells were grown in M9 minimal media supplemented with 0.34% (w/v) sodium succinate (Sigma-Aldrich S2378) at 30 °C with shaking at 220 rpm. Starter cultures were grown for 12 to 72 h, diluted ≥1,000-fold, and then grown for several days to early exponential phase (OD_{600} = 0.095 to 0.105) prior to imaging.

Glass slides (Fisher 12-544-1) and #1.5 glass coverslips (Fisher 12544A or VWR 16004-344) were cleaned by three rounds of sonication (15 min each) in 100% ethanol, ~80% ethanol, and Milli-Q water. Cleaned slides and coverslips were stored in Milli-Q water until the day of imaging and dried with nitrogen gas immediately before use.

Agarose pads (1% w/v) were prepared by dissolving agarose (Invitrogen 16500-100) in the same type of medium used for cell growth. Molten agarose was solidified between two cleaned glass slides to ensure a flat surface. Cells were applied to the agarose pad, and excess liquid was removed by air drying for ~1 min. A cleaned coverslip was then placed on top and sealed with VALAP. Samples were imaged immediately. When rifampicin-treated cells were imaged, 200 μg/mL of rifampicin was added to the molten agarose kept at 60 °C and solidified right away.

Drug Treatment. Cells were treated with rifampicin (400 μg/mL, Sigma R3501) for 15 min in the liquid culture with shaking at the growth temperature.

Single-Molecule Imaging. Bright-field and single-molecule mEos3.2 fluorescence imaging was performed on a Nikon Ti-2 microscope as previously described (20). Imaging conditions, including hardware configuration and laser settings, were identical to those used in our previous study (20). Briefly, a quasi-TIRF illumination geometry was used, with a 405 nm laser for photoconversion of mEos3.2 and a 561 nm laser for excitation of the photoconverted state. Movies were acquired for 3 min on an EM CCD camera with an effective pixel size of 160 nm at the sample plane, using a 21.7-ms exposure and a 21.7-ms frame interval under continuous illumination. Single-molecule positions were obtained by

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subpixel localization of the diffraction-limited fluorescence spots. Each strain was imaged in three biological replicates, with 6 to 16 movies collected per replicate.

***lacZ* Induction, RNA Extraction, and RT-qPCR.** Pulsed induction of the *lacZ* reporter, RNA extraction, and RT-qPCR were performed as previously described (32). Cells were grown in M9 minimal medium supplemented with 0.2% (v/v) glycerol and CAAT at 30 °C to early exponential phase (OD_{600} = 0.2). Following induction with 0.2 mM IPTG at t = 0, 0.4 mL of culture was withdrawn every 20 to 60 s and immediately mixed with an equal volume of precooled RNAlater Stabilization Solution (Life Technologies). 500 mM glucose was added to the culture at t = 75 s to repress the transcription. After mixing with RNAlater, samples were incubated on ice for 20 min and then centrifuged. Cell pellets were resuspended in lysozyme solution [10 mg/mL of lysozyme in 10 mM Tris-HCl (pH 8) with 1 mM EDTA]. Total RNA was extracted using the PureLink RNA Mini Kit (Life Technologies) with on-column DNase treatment according to the manufacturer's instructions. RT-qPCR was performed using KAPA SYBR FAST qPCR Master Mix (KAPA Biosystems) on a Bio-Rad CFX Connect system. Primer sequences are provided in *SI Appendix*, Table S4.

Relative RNA levels were calculated as fold change relative to baseline (preinduction samples) using the $2^{-\Delta Ct}$ method, where Ct denotes the cycle threshold. RNA degradation rates were obtained by fitting log-transformed RNA levels within defined time windows. Raw RT-qPCR data are provided in Dataset S1.

Data Analysis.

Single-molecule tracking from live cells. As previously described (20), data were analyzed in Matlab using Oufti (41) for cell detection from bright-field images, u-track (42) for fluorescence spot detection and linking into tracks, a custom-built Matlab code called spotNorm (20) for finding a fluorescence spot's normalized position in a cell, and custom-built Python code for calculating the membrane-binding percentage (MB%).

Oufti. Cell outlines were computed from bright field images using Oufti software developed by Jacobs-Wagner Lab (41). The parameters used for Oufti for mEos3.2 labeled cells are the same as described previously (20).

spotNorm. Normalized spot positions within cells were calculated as previously described (20). Briefly, spot locations were normalized along the minor axis (xNorm) and the major axis (yNorm) of each cell. xNorm histograms were constructed using the first four spots per track located within the cylindrical region of the cell. The SEM shown for the xNorm histograms (e.g., Fig. 2 *C–I*) was estimated by bootstrapping across cells, but it is very small, often smaller than the data line width.

u-track for mEos3.2 tracking. Fluorescence spots were detected and compiled into tracks using u-track software developed by Danuser Lab and Jaqaman Lab (42). For mEos3.2 tracking, we used the same parameters as our previous paper (20). A notable parameter in u-track is the max pixel linking distance in the Brownian and directed motion models specifically called the upper bound for the Brownian search radius (in pixels). We set this parameter to 1 to 3 pixels depending on the preliminary diffusion coefficient based on trajectories identified using u-tracks' default 5-pixel maximum linking distance, D_{5pix}. We previously found that the max linking distance of 5 pixels can create artificial large random jumps in our dataset that falsely creates a larger diffusion coefficient. To avoid this error, we set the max linking distance radius as ceiling(2*sqrt(4D_{5pix}Δt)), where Δt is the frame time (0.0217 s). Alternatively, we can use 1 pixel if $D_{5pix} < 0.0737$ μm^2/s and 2 pixels if 0.0737 μm^2/s < $D_{5pix} < 0.295$ μm^2/s, etc. For proteins with expected slow and fast states (PNPase and enolase), we added one more pixel to prevent cutting out faster diffusing particles. In other words, we used 2 pixels if $D_{5pix} < 0.0737$ μm^2/s, 3 pixels if 0.0737 μm^2/s < $D_{5pix} < 0.295$ μm^2/s, etc. Effectively, the max pixel values we used are as follows:

- 5 pixels: Eno (rif)
- 3 pixels: β-Gal, β-Gal (rif), β-Gal (succ), RhlB (rif), Eno, Eno (succ)
- 2 pixels: LacY, LacY (rif), LacY (succ), RNE (succ), RhlB (succ), PNP, PNP (rif), PNP (succ)
- 1 pixel: RNE, RNE (rif), RhlB

Mobility analysis by per-trajectory MSD and squared displacements.

Diffusion by MSD. Tracks containing at least 12 consecutive localizations were used for diffusion analysis as previously described (20). Time-averaged (TA) MSD was calculated according to Eq. **1**, and the diffusion coefficient (D) was calculated by a linear fit using Eq. **2** (24):

$$MSD(\tau) = \frac{1}{N_\tau}\sum_{i=1}^{N_\tau}\left[\vec{r}\left(t_i\right)-\vec{r}(t_i+\tau)\right]^2, \quad [1]$$

$$MSD(\tau) = 4D\tau + b, \text{ where } b = -\frac{4Dt_E}{3} + 4\sigma^2. \quad [2]$$

In Eq. **1**, $\vec{r}\left(t_i\right)$ is the 2D vector location of a particle at time t_i, τ is the lag time, and N_τ is the number of frames averaged in a track for that lag time. In Eq. **2**, t_E is the exposure time (21.7 ms), and σ is the static localization error. The y-intercept, b, is composed of dynamic localization error caused by motion averaging during the finite exposure (t_E) and static localization error arising from finite pixel sampling, photon statistics, background fluorescence, and camera noise (24). We fit the first three time points of TA MSD with Eq. **2** to get D. This ensures that the fit is done on short time delays (the first 25% of the minimum track length) to avoid MSD at large time intervals with low statistics (43, 44). This also minimizes, although it does not eliminate, effects of spatial confinement imposed by the finite cell geometry.

The fitted D values represent apparent diffusion coefficients. Although localization uncertainty and finite-exposure motion averaging are accounted for to first order through the intercept in Eq. **2**, the resulting D can also be influenced by confinement within the small cell and by projection of motion along the curved membrane onto the imaging plane for membrane-associated molecules. The requirement of at least 12 consecutive localizations may preferentially retain slower molecules because fast molecules are more likely to leave the detection volume or fail trajectory linking (*SI Appendix*, *Supplementary Note 2*). The same acquisition, track-length, and fitting criteria were applied to all strains and conditions. Accordingly, the reported D values are interpreted as a comparative measure of molecular mobility under matched imaging and analysis conditions rather than as an intrinsic, unconstrained three-dimensional diffusion coefficient.

Once an apparent diffusion coefficient was obtained for each trajectory from MSD fitting, the distribution of D values was represented as a probability histogram and fit with a mixture of Gaussian functions:

$$P(D) = c\sum_{i=1}^{N} p_i \frac{1}{\sqrt{2\pi}\sigma_i}\exp\left[-\frac{\left(D-\mu_i\right)^2}{2\sigma_i^2}\right], \quad [3]$$

where N denotes the number of fitted mobility populations, p_i is the fraction of population i, μ_i and σ_i are its mean and SD, respectively, and c is a scaling factor accounting for the histogram bin width. The fitted means μ_i are denoted D_1, D_2, and D_3 in the main text. The number of Gaussian components (N) was increased sequentially from one to three, and the simplest model that adequately reproduced the overall distribution, including the higher-mobility tail ($R^2 > 0.97$), was selected.

For a two-population fit, the fractions were parameterized as $p_1 = f_1$ and $p_2 = 1-f_1$. For a three-population fit, they were parameterized as

$$p_1 = f_1,\ \ p_2 = \left(1-f_1\right)f_2,\ \ p_3 = \left(1-f_1\right)\left(1-f_2\right), \quad [4]$$

with $0 \le f_1, f_2 \le 1$, thereby constraining all population fractions to be nonnegative and sum to unity.

Diffusion by one-frame squared displacements. As a complementary analysis that included shorter trajectories, we analyzed one-frame squared displacements from all trajectories containing at least four consecutive localizations. Three one-frame squared displacements (r^2) were included from each trajectory so that all trajectories contributed equally, independent of track length. The resulting values were combined to construct the empirical survival distribution, or complementary cumulative distribution function, CCDF, $1 - F\left(r^2\right)$.

For ideal two-dimensional Brownian diffusion, the survival distribution of the one-frame squared displacement is exponential:

$$1 - F\left(r^2\right) = \exp\left(-\frac{r^2}{4D\Delta t}\right). \quad [5]$$

To resolve multiple mobility populations, the empirical CCDF was fit with a mixture of exponential functions:

$$1 - F\left(r^2\right) = \sum_{i=1}^{N} p_i\ \exp\left(-\frac{r^2}{\lambda_i}\right), \quad [6]$$

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where $N = 2$ or 3 is the number of fitted mobility populations, p_i is the fraction of population i, and λ_i is the characteristic squared-displacement scale. The apparent diffusion coefficient of each population was calculated as

$$D_i = \frac{\lambda_i}{4\Delta t}, \quad [7]$$

with $\Delta t = 21.7$ ms. The CCDF was first fit with a two-population model, and a third component was added only when the two-population model did not provide an adequate fit to the data. For the three-population model, the population fractions were constrained using the parameterization described above to ensure that all fractions were nonnegative and summed to unity. At large squared displacements, the empirical CCDF deviated downward from the expected exponential behavior, likely because of limited sampling of rare large-displacement events. Therefore, fitting was restricted to data points corresponding to displacements below half of the max linking distance.

Fitting and uncertainty. Gaussian-mixture and exponential-mixture models were fit by nonlinear least-squares optimization in MATLAB using custom fittype models. Goodness of fit was evaluated using the coefficient of determination, R^2. The 95% CI of the fitted diffusion coefficients and population fractions were obtained from the nonlinear least-squares fits. Fit results are summarized in *SI Appendix*, Table S5 and Dataset S2.

MB% quantification from xNorm histograms. Membrane-bound fraction (MB%) was estimated from xNorm histograms using the modeling framework established in our previous study (20). Briefly, bacterial cells were approximated as cylinders, and molecules were assumed to be homogeneously distributed either on the cell surface (membrane-associated population) or within the cell interior (cytoplasmic population). For each case, the expected xNorm distribution was calculated from the marginal distribution of molecular positions along the short cell axis. The observed distribution was then modeled as a mixture of membrane and cytoplasmic populations, with the membrane-associated fraction corresponding to MB%.

To account for imaging effects, the model incorporated localization uncertainty and finite axial detection range, as described previously (20). Cell width measured from bright-field images was used to scale the model to each dataset, and xNorm values were fit in normalized coordinates. Experimental histograms of absolute xNorm values were fit to the model using Markov chain Monte Carlo (MCMC) sampling with uniform priors on all fitted parameters, including MB%, localization error, dilation factor, and focal cutoff. The posterior estimate of MB% was taken as the membrane-bound fraction. Full mathematical derivation, implementation details, and parameter definitions are provided in ref. 20.

Data, Materials, and Software Availability. All code used for data analysis is available in Github (sjkimlab/2025_RNaseE) (45). Processed microscopy data are available on Zenodo (10.5281/zenodo.20384234) (46) and RT-qPCR data are included in the manuscript as Dataset S1.

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